Visarjan Ghat Watch

Master Project Document

One-line description: A pre-festival water-quality risk briefing tool that helps Bhopal Municipal Corporation officers prioritize testing, cleaning and aeration at idol-immersion ghats using traceable historical MPPCB readings.

Document field	Value
Document purpose	Master reference for understanding, designing, building, validating, operating and extending the project
Project type	Public-interest software / data-analysis tool
Current stage	Specification defined; implementation status is not verifiable from the supplied documents
Document version	1.0
Document date	11 October 2026
Project owner/team	Project team; member names were not included in the supplied sources
Primary user	Bhopal Municipal Corporation (BMC) officer planning Durga visarjan at the ghats
Planning context	Claude Impact Lab x Cerebras Cafe Compute, Bhopal - Track 2

> ***Source and status rule:*** *This document consolidates the supplied Visarjan Ghat Watch: Solution Architecture & Project Specification and Track 2: Visarjan Ghat Watch - Participant Guide. Features are described as requirements or plans unless the supplied documents explicitly establish that they have been implemented. No implementation result, final ranking, deployment URL, or measured user outcome is assumed.*

Document map

0. Project overview and scope
1. Background and problem definition
2. Research and discovery
3. Goals, requirements and success criteria
4. Proposed solution and product design
5. System architecture and design
6. Technical approach and core logic
7. Technology stack and tools
8. Implementation plan
9. Challenges, risks and mitigation
10. Testing, validation and quality assurance
11. Deployment, operations and maintenance
12. Impact, value and differentiation
13. Feasibility, viability and sustainability
14. Team, responsibilities and governance
15. Future roadmap and expansion
16. Documentation, references and glossary
17. Open questions and action items

0. Project Overview

0.1 Elevator pitch

Visarjan Ghat Watch turns historical water-quality measurements that are buried in long MPPCB reports into a ranked, evidence-backed briefing for the BMC officer planning Durga visarjan. It compares relevant readings with CPCB Class B criteria, keeps missing measurements visibly missing, traces displayed figures to their report pages, and prepares a short Hindi and English brief for operational planning.

0.2 Problem being addressed

MPPCB reports water-quality readings around Ganesh and Durga immersion at three stages: before immersion, during immersion and about one week afterwards. The supplied guide says these findings are published in reports roughly 87-118 pages long, while the solution document describes reports of about 90-120 pages. The results are not conveniently aligned against the relevant criteria or compared across years for planning purposes.

The supplied documents highlight these findings for Bhopal ghats across 2023-2025:

- Dissolved oxygen (DO) was reported as **0 mg/l four times** at Bairagarh (Sant Hirdaram) kund across 2023 and 2025.
- **42 of 116 DO readings** were below the CPCB Class B threshold of 5 mg/l.
- **100 of 112 BOD readings** were above the CPCB Class B limit of 3 mg/l.

These are figures stated in the supplied project guide and solution specification. The build must reproduce the headline counts from the provided data before the dashboard relies on them.

0.3 Proposed solution

The project is designed to provide one focused workflow:

1. Rank seven Bhopal ghats by evidence-based risk, with a plain-language reason for each rank.
2. Let the officer inspect the measurements behind a conclusion, including year, festival, sampling phase, value, unit, flag and source PDF page.
3. Show data gaps directly so lack of sampling is not mistaken for good water quality.
4. Generate a short, shareable bilingual briefing with suggested testing, aeration and cleaning priorities, plus a clear limitations statement.

0.4 Target user and primary use case

Primary user: The BMC officer responsible for planning Durga visarjan at the ghats.

Primary use case: Before the festival, the officer reviews the ranked ghats, inspects the evidence and gaps, then uses the briefing to help prioritize where additional testing, cleaning or aeration should be considered.

The tool supports the officer's decision; it does not autonomously dispatch teams or make official regulatory or health determinations.

0.5 Core differentiator

Trustworthy, auditable briefing rather than an opaque AI answer. Python rules calculate metrics and risk tiers deterministically. An optional language model may draft explanations and Hindi text from a verified facts table, but it must not originate numerical claims. A checker verifies each number against the facts table and a phrase list blocks unsupported health claims.

0.6 Project objectives

- Make the historical water-quality evidence usable for a time-sensitive pre-festival decision.
- Compare relevant readings against clearly stated CPCB Class B criteria.
- Make both severity and confidence in the available evidence visible.
- Preserve traceability from displayed values to source report pages.
- Provide a short Hindi and English operational brief.
- Be usable from a phone through a link, without installation or a user account.
- State data gaps and analytical limits honestly.

0.7 Project scope

In scope for the specified MVP

- The 1,524-row Bhopal data file covering seven ghats and the 2023-2025 period.
- Explicit treatment of missing values and BDL/ND flags.
- Rule-based metrics against the specified CPCB Class B thresholds for DO, BOD and pH.
- A ranked view of seven ghats, evidence-strength information and a phase view.
- A readings/evidence table and a panel explaining missing years and phases.
- A Hindi and English one-page brief, delivered as copy-ready text and a printable PDF.

- Automated numeric verification and checks against prohibited health claims.

Out of scope for the initial version

- Health-effect predictions or medical advice.
- Applying Class B limits to chromium, lead, zinc or copper when the selected criteria do not specify Class B limits for these metals.
- A general portal, user accounts, login system or multi-user administration.
- Live connections to government systems or live water sensors.
- Treating records for the old ghats as if they represented the four new BMC ghats.

Assumptions and constraints

- The provided public datasets and source reports are the initial data sources.
- The tool is a decision-support briefing, not a replacement for fresh sampling or the officer's judgment.
- The guide's source-page field can be used to trace a reading into the corresponding source PDF.
- The Class B comparison is used because organised outdoor bathing is identified in the source documents as the closest designated use to immersion; this rationale must be stated in the product.
- No live government portal is to be logged into, scraped or written to during the event build.
- The event guide requires an open-source repository and says no API keys or secrets should be committed.

---

1. Background and Problem Definition

1.1 Background

The MP Pollution Control Board (MPPCB) monitors water quality around Ganesh and Durga idol immersion. The readings cover a number of parameters and are recorded for phases before, during and after immersion. The source documents explain that a reader must interpret abbreviated parameter names, assumed units, missing markers and differing site labels, then compare values against applicable water-quality criteria.

The project guide also describes a planning context in which the National Green Tribunal (NGT) has banned immersion in the Upper and Lower Lakes and BMC is building dedicated ghats. The immediate planning milestone identified in the source documents is Durga visarjan on 20 October 2026.

1.2 Problem statement

> *A BMC officer planning Durga visarjan needs to decide where additional water testing, cleaning and aeration deserve attention, but the historic evidence is fragmented across lengthy reports and incomplete sampling records. Without a consistent comparison to stated limits and a visible account of missing data, the officer must manually reconstruct trends and may not know how strong or weak the evidence is for each ghat.*

1.3 Current workflow described by the sources

Current workflow: MPPCB PDF reports → manual search for readings, years and phases → manual comparison across years → officer relies heavily on memory to prioritize action.

Proposed workflow:

1. Load the historical CSV, source reports and CPCB criteria.
2. Clean values while preserving the difference between missing, BDL and ND.
3. Calculate per-ghat metrics and evidence coverage.
4. Create a traceable facts table and deterministic ranking.
5. Draft plain-language reasons and Hindi text from verified facts.
6. Verify generated numbers and screen for unsupported health claims.
7. Publish the static dashboard, evidence view and bilingual brief.

1.4 Root-cause analysis

Observed problem	Underlying cause described in the sources	Consequence for the user
Values are hard to compare across years	Readings are embedded in lengthy reports and not aligned in a single decision view	A cross-year pattern is difficult to review quickly
A blank may be misinterpreted	Reports use blanks, underscores, dashes, asterisks and “not sampled” markers	Missing evidence could incorrectly appear reassuring
BDL or ND may be misread	These labels mean below detection limit / not detected, not zero	The analysis could distort the measurement's meaning
Some ghats have incomplete phases or years	Sampling coverage differs; some ghats were sampled only during immersion and some recent tables are absent	Evidence strength differs across ghats
Site naming is ambiguous	Ganesh and Durga reports use different names for Rani Kamlapati ghat / Kamla Park kund	Combining records without a warning could join different places

Observed problem	Underlying cause described in the sources	Consequence for the user
AI-generated prose could introduce unsupported numbers	Language models can generate text not present in the evidence	An unverified brief could damage trust in the ranking

1.5 Stakeholders

Stakeholder	Relationship to the project	Need or responsibility
BMC planning officer	Primary user	Review risk, evidence gaps and suggested operational priorities
MPPCB	Source of historical monitoring reports	Readings and source-page evidence underpin the analysis
CPCB	Source of water-quality criteria	Defines the comparison thresholds referenced by the project
BMC teams responsible for field actions	Potential downstream users of the brief	Receive clearer priorities for testing, cleaning and aeration
Project developers and maintainers	Build and maintain the tool	Preserve data lineage, reproducibility and clear limitations
Government department / future adopters	Potential recipient after the event	Need readable code, a clear README, licence and run instructions

1.6 Why this matters

The stated value is faster, more transparent planning from existing public evidence. The product should help the officer distinguish two different things: how badly readings breach selected limits, and how complete or incomplete the evidence is. It must not turn historical comparisons into unsupported claims about human health or assert that a ghat is safe solely because data is missing.

2. Research and Discovery

2.1 Domain rules and terminology

The solution specification selects **CPCB Class B - outdoor bathing (organised)** as the comparison standard because it is considered the closest designated use to immersion among the listed classes. The interface should explain this choice rather than imply that immersion has a bespoke standard in the dataset.

Parameter	Unit in the guide	Class B threshold specified for the project	How the MVP treats it
Dissolved oxygen (DO)	mg/l	5 mg/l or more	Flag readings below 5 mg/l as breaches
Biochemical oxygen demand (BOD)	mg/l	3 mg/l or less	Flag readings above 3 mg/l as breaches
pH	none	6.5 to 8.5	Flag readings outside the interval as breaches
Total coliform	MPN/100 ml	500 or less	Part of Class B criteria, but absent from the supplied immersion dataset; state as a limitation
Chromium, lead, zinc, copper	mg/l	No Class B limit specified in the source criteria	Show readings only; do not score against a Class B limit
Suspended solids	mg/l	No Class B limit specified in the source criteria	Do not compare across years where the relevant Durga tables are absent
Other parameters (COD, solids, turbidity, conductivity, etc.)	Parameter-specific	No Class B limit specified for them in the source table	Retain as context where useful; do not invent a Class B pass/fail threshold

Important interpretation rules

- A blank, `_`, `-`, `*`, `**` or `not sampled` means missing/not sampled, never “clean”.
- BDL means below detection limit; ND means not detected. The guide describes these as very low, not zero and not missing.
- Preserve the printed form (`value_as_printed`) as well as the normalized value and flag where available.
- The report does not print units consistently; `unit_assumed` records the standard unit associated with each test.
- Keep Ganesh and Durga readings separate in the evidence view. Durga is weighted more heavily for the planned Durga event; Ganesh is supporting context.

2.2 Data inventory

File or source	Size / period	Intended role	Known caveat from the guide
data/ bhopal_ghats_2023_2025 .csv	1,524 rows; seven Bhopal ghats; 2023-2025	Primary MVP dataset	Coverage varies by ghat, year, festival and phase
data/ sample_bhopal_2025.csv	453 rows	Quick development and smoke-test subset	Contains only 2025 records from the primary file
data/ mp_immersion_sites_202 3_2025.csv	21,600 rows; about 100 sites per festival across Madhya Pradesh	Future statewide expansion	Less checked; records marked needs_check may not have readable site names
data/ cpcb_water_quality_criteri a.csv	19 rows	Structured source for classes A-E criteria	Use the criteria relevant to the chosen class; do not infer missing limits
MPPCB Ganesh and Durga Visarjan Report 2023	87 pages, as listed in the guide	Primary historical evidence	Use source_page to locate a reading
MPPCB Ganesh and Durga Visarjan Report 2024	118 pages, as listed in the guide	Primary historical evidence	Use source_page to locate a reading
MPPCB Ganesh and Durga Visarjan Report 2025	92 pages, as listed in the guide	Primary historical evidence	Use source_page to locate a reading
Upper/Lower Lake water-quality reports	Multiple periods listed in the guide	Context and possible future enhancement	Not part of the core seven-ghat scoring unless explicitly incorporated and documented

The file names above are taken from the participant guide; verify exact paths in the supplied repository before running the pipeline.

2.3 Key fields in the immersion dataset

Field	Meaning and use
year	Report year
festival	Ganesh or Durga
site	Standardized Bhopal ghat name in the Bhopal file
water_body	Lake or kund associated with a Bhopal site
phase	pre, during or after
parameter	Standardized parameter name
value	Numeric reading where supplied; can be empty where the report has no number
unit_assumed	Standard unit associated with the test
flag	Data status such as BDL or not sampled
value_as_printed	Original text as presented in the report
parameter_as_printed	Parameter label exactly as printed
site_as_printed	Site name exactly as printed
site_match_confidence	Confidence in standardizing the site name; used in the Bhopal file
needs_check	Statewide data flag for a site name that needs review
source_page	PDF page where the measurement can be checked

2.4 The seven ghats in the MVP

Standard name	Water body	Site-mapping note
Bairagarh (Sant Hirdaram) kund	Upper Lake	High match confidence; no Durga 2025 table in the report
Khatlapura ghat	Lower Lake	High match confidence
Prempura ghat	Upper Lake	High match confidence
Rani Kamlapati ghat / Kamla Park kund	Lower Lake	Medium confidence; Ganesh tables use “Kamla Park kund” while Durga tables use “Lower Lake ghat”; may be two places
Hathaikheda dam / kund	Hathaikheda dam	High match confidence

Standard name	Water body	Site-mapping note
Malikhedi kund	Kund	High match confidence; no Durga 2025 table in the report
Shahpura lake ghat	Shahpura Lake	High match confidence

The ranking must not overstate certainty where a site identity or recent sampling coverage is unclear.

2.5 Existing workflow and opportunity

The baseline described by the sources is manual interpretation of lengthy reports, with decisions relying substantially on memory. The project opportunity is not to replace sampling; it is to structure existing evidence, show its limits, make source verification practical, and turn data gaps into explicit prompts for additional testing.

2.6 Evidence quality and research conclusions

The guide says the Bhopal CSV has been spot-checked against the source PDFs. The solution specification additionally requires the build's unit test to reproduce the headline counts and recommends manually spot-checking five to ten readings against the PDFs through `source_page`. These should be treated as verification tasks for each delivered build, not as evidence that the current implementation has already passed them.

---

3. Goals, Requirements and Success Criteria

3.1 Goals

1. Provide an interpretable ranking of the seven specified ghats.
2. Make the evidence behind each rank inspectable.
3. Separate severity from evidence strength.
4. Generate a short bilingual pre-festival brief.
5. Prevent missing data, unsupported AI-generated numbers and unsupported health claims from being presented as facts.

3.2 Functional requirements

Priority labels: **Must** = described as MVP; **Should** = listed differentiator planned after the MVP; **Could** = stretch goal or future enhancement. These labels describe the supplied specification, not implementation completion.

ID	Requirement	Priority	Acceptance condition
FR-01	Load the 1,524-row Bhopal dataset	Must	Pipeline reads the expected dataset and reports row/data issues instead of silently dropping them
FR-02	Normalize missing markers separately from BDL/ND	Must	Blank, _, -, *, **, and not sampled remain missing; BDL/ND are not converted to zero
FR-03	Compare DO, BOD and pH to CPCB Class B thresholds	Must	Thresholds are explicit and boundary values are handled consistently
FR-04	Compute per-ghat metrics	Must	Output includes sampled readings, breaches, worst value, year trend and pre/during/after pattern where data exists
FR-05	Weight Durga readings more heavily than Ganesh context	Must	Weighting is documented and reproducible; exact weights must be defined in implementation documentation if numeric weights are used
FR-06	Rank seven ghats and explain each rank	Must	Each ghat has a risk tier, plain-language reason, evidence-strength indicator and phase summary
FR-07	Separate risk severity from evidence strength	Must	The interface displays the two axes independently; sparse data never produces an implicitly safe impression
FR-08	Provide a reading-level evidence view	Must	Each record view includes year, festival, phase, value, unit, flag and source page
FR-09	Display missing phases and years	Must	A “What we don't know” panel shows gaps for each ghat; missing is visually distinct from BDL
FR-10	Flag the ambiguous Rani Kamlapati / Kamla Park site mapping	Must	Ganesh and Durga labels stay separate and a “may be two places” warning is visible
FR-11	Generate a Hindi and English brief	Must	Brief contains top-risk ghats, three suggested action categories and a data-limitations box
FR-12	Export/copy the brief	Must	Copy-ready text and a printable PDF are available

ID	Requirement	Priority	Acceptance condition
FR-13	Verify every number in generated explanatory text	Must / core trust control	Every numeric claim in generated reasons and the Hindi brief is checked against the facts table; unverified claims are blocked or flagged
FR-14	Block unsupported health-effect claims	Must / core trust control	Banned-phrase checks run over generated prose and the brief states that it reports against limits only
FR-15	Produce a second ranking for where new testing is most valuable	Should	Test-Me-First ranks severity and uncertainty, and gaps can increase a site's testing priority
FR-16	Support a tamper-test demonstration	Should	A changed number is caught by the numeric checker; verification status is visible
FR-17	Recompute rankings under alternative assumptions	Should	User can compare Class B/C/D, Durga-only vs both festivals, and merged vs separate Rani Kamlapati records
FR-18	Show source-report page receipts and evidence-linked QR codes	Could	Clicking a value can display its cited PDF page and the brief can link to an evidence view
FR-19	Export a WhatsApp-friendly image card and spoken Hindi summary	Could	English/Hindi 1080-pixel card is available and browser speech synthesis can read the brief

3.3 Non-functional requirements

Area	Requirement
Trust and traceability	Displayed metrics must be reproducible from source data and rules, with source-page references retained
Data semantics	Missing, BDL and ND states must remain distinguishable through parsing, calculations, visualizations and exports
Explainability	The user should be able to understand why a ghat is ranked where it is and how much evidence supports the ranking
Safety of claims	Report readings against limits; do not make unsupported health claims or claim that missing evidence implies safety
Usability	A BMC officer should be able to open a link on a phone without installing an app or signing in
Maintainability	Readable code, reproducible build, short run instructions and open-source repository
Runtime simplicity	Static site is preferred; no runtime API or database is required by the specified architecture
Bilingual output	Hindi text and PDF output must render correctly; bundle and test Noto Sans Devanagari early
Secret handling	API keys remain in environment variables and are never committed to the repository
Scope discipline	One workflow, one primary user, one outcome; no portal-style feature expansion in the MVP

3.4 Success metrics and acceptance targets

These are target checks for a build; passing status is not established by the supplied documents.

Metric / check	Target	Verification method
DO headline count	42 of 116 readings below 5 mg/l	Automated test against the Bhopal dataset
BOD headline count	100 of 112 readings above 3 mg/l	Automated test against the Bhopal dataset
Source validation	Manually spot-check 5-10 readings	Open the cited report pages and compare the original value, phase and site
Ghat coverage	Exactly seven specified MVP ghats represented	Dataset and output audit
Numeric claims	Every displayed number in generated reasons / brief exists in the facts table	Automated number-auditor output and tamper test
Health claims	Zero unsupported health-effect claims	Phrase audit and manual review
Missing-data semantics	No missing value shown as clean, zero or safe by default	Unit tests plus UI inspection of missing/BDL examples
Traceability	Readings in the evidence drawer include the requested source context	Sample data-to-UI verification
Bilingual brief	Hindi and English brief renders and remains readable when printed	Browser and PDF rendering checks
Site ambiguity	Rani Kamlapati / Kamla Park ambiguity is visible	UI acceptance check

3.5 Constraints and dependencies

The build depends on the supplied CSV files, the CPCB criteria file, report PDFs, Python and pandas, and a static-site workflow. If an LLM is used for prose, access to the selected inference provider is a build-time dependency, not a runtime dependency. Total coliform data is not available in the primary dataset according to the source guide.

4. Proposed Solution and Product Design

4.1 Core user experience

The product should open directly to a ranked overview, not a generic landing page or portal. The officer should see where risk indicators are concerning, why a ghat is positioned at that level, how much evidence exists and which important data gaps remain. Selecting a ghat should expose its evidence and the underlying source pages. A short brief should be available to copy or print.

4.2 MVP features

F1. Data and risk engine

Loads the Bhopal CSV, applies the missing-data rules, compares the supported parameters against Class B criteria, calculates ghat-level metrics, gives Durga data greater relevance for the planned Durga event, and outputs facts and source-page references. Metals are shown as readings only; suspended solids are not compared across years where the relevant Durga tables are absent.

F2. Ranked ghat view

Seven ghat cards, each with a risk tier, a one-sentence reason, evidence-strength badge and mini phase chart. Severity and evidence strength must be shown as two different dimensions. A ghat with few readings must not appear safe merely because few breaches were observed.

F3. Evidence and gaps drawer

A reading table with year, festival, phase, value, unit, flag and source page, plus a “What we don't know” panel listing missing years and phases. “Not sampled” and BDL use different visual treatments. The Rani Kamlapati / Kamla Park naming ambiguity must remain visible.

F4. Bilingual briefing

A one-page Hindi and English summary of the top-risk ghats, three action categories (testing, aeration and cleaning priorities) and limitations of the data. Provide copy-ready text and a printable PDF.

4.3 Differentiators and stretch goals

Feature	Priority in supplied specification	Intended behavior
Test-Me-First	First differentiator	Produces a second ranking for where a new test is most valuable before 20 October, based on severity x uncertainty. Bad readings plus missing phases or no recent table should increase testing priority.
Tamper Test / live number auditor	Second differentiator	Checks every numeric claim against the facts table and blocks health-claim phrases. A visible status badge summarizes the audit. A demonstration edit should be caught by the checker.
Rank-stability dial	Third differentiator	Recalculates results for alternate assumptions: Class B vs C vs D; Durga-only vs both festivals; and with/without the potentially ambiguous site merge. It distinguishes robustly high-risk results from assumption-dependent results.
Receipts	Stretch goal	Renders the cited PDF page (whole page or relevant crop) when a value is clicked; the brief can include a QR code to the evidence view.
Pocket Briefing	Stretch goal	Exports a 1080-pixel Hindi and English shareable card and offers “Listen in Hindi” using browser speech synthesis. Offline behavior is a stated target for the static site once opened.

4.4 User journey

1. The officer opens the static link.
2. The officer reviews the seven-ghat risk ranking.
3. The officer checks the reason, severity and evidence strength.
4. The officer opens the readings and missing-data details.
5. The officer follows a number to its year, festival, phase and report page.
6. The officer reviews the Hindi-English brief and its limitations.
7. The officer uses the information to plan testing, cleaning and aeration.

4.5 Primary use cases

Use case	User action	Expected system response
Compare ghats before Durga visarjan	Open the dashboard	Show all seven ghats, ranking reasons, severity and evidence strength
Validate a concerning value	Select a ghat and a reading	Show year, festival, phase, value, unit, flag and source page
Understand incomplete evidence	Open data-gap details	Show missing years/phases clearly and avoid interpreting them as clean results
Prepare an operational brief	Open brief view/export	Provide Hindi and English copy-ready content and a printable PDF
Evaluate where a new test is most useful	Use Test-Me-First if implemented	Prioritize severity together with uncertainty and data gaps
Check reliability of AI-generated explanation	Inspect verification status / run tamper test	Show whether numeric claims are grounded in the facts table and flag an edited unsupported number

4.6 Product boundaries

The system calculates and presents historical evidence. It does not conduct sampling, validate real-time water conditions, issue regulatory determinations, predict health impacts, send crews, or establish that a site is safe. Suggestions about testing, aeration and cleaning remain decision support for the responsible officer.

---

5. System Architecture and Design

5.1 Recommended architecture

The specified architecture performs all data processing at build time. The browser reads generated static files; no backend, database, REST API or WebSocket connection is required at runtime.

Stage	Processing / output
Data sources	Bhopal CSV (1,524 rows), CPCB criteria and source-page metadata
Batch pipeline	Python 3 + pandas via build.py
Cleaning	Preserve numeric values, BDL, ND and missing states separately
Metrics and ranking	Per-ghat metrics, deterministic severity tiers and separate evidence-strength information
Facts and provenance	Verified facts table with source-page references
Optional language drafting	LLM drafts reasons and Hindi text from the facts table; no new numbers are permitted
Verification	Python number checker plus health-claim phrase list
Generated artifacts	ranking.json, evidence.json, brief text and printable PDF
Delivery	Static website / hosted link

5.2 Component breakdown

Component	Responsibility	Input	Output
Data loader	Read project CSVs and criteria	Bhopal CSV, criteria CSV	Loaded records with source fields
Cleaning and semantic normalization	Preserve data meaning and separate missing from BDL/ND	Raw values and flags	Normalized records with status retained
Metrics and rule engine	Calculate sample counts, breaches, worst values, trends and phase patterns	Normalized records + Class B thresholds	Per-ghat metrics and deterministic severity tier inputs

Component	Responsibility	Input	Output
Evidence/facts builder	Pair calculated facts with report provenance	Metrics and source-page fields	Facts table, evidence JSON
Ranking module	Produce seven-ghat ordering and separate severity/evidence indicators	Metrics and documented weighting	ranking.json and ranking metadata
Optional language-model drafting	Convert approved facts into plain-language reasons and Hindi brief wording	Verified facts table	Draft explanations; no new numbers permitted
Verification module	Check numeric claims and prohibited health language	Draft text + facts table + phrase list	Verified or flagged output and audit status
Frontend	Display ranked ghats, evidence, gaps, charts and brief	Generated JSON / static files	Phone-friendly static user experience
Brief exporter	Produce readable bilingual text and PDF	Approved brief + bundled Devanagari font	Copy-ready text and printable PDF
Optional receipt renderer	Render source PDF pages at build time	source_page and source PDFs	Evidence-page images

5.3 Data flow

1. `build.py` reads the Bhopal CSV and applies missing-value rules while preserving the distinction between missing, BDL and ND.
2. The engine compares supported readings to CPCB Class B criteria and calculates ghat-level metrics.
3. The ranking logic produces severity and evidence-strength information separately; Durga readings receive greater weight for the planned Durga event.
4. A facts table stores values, calculated metrics and source-page links. This table is the approved numeric basis for explanations.
5. An optional LLM drafts reasons and Hindi text from that table. It is not allowed to originate numeric facts.
6. A number checker compares generated numbers with the facts table, while a phrase list screens against unsupported health claims.
7. The build emits static JSON and brief files, and the website displays them without a live backend.

The source specification does not define every internal formula or threshold for the overall risk tier. Those implementation choices must be explicitly documented, reproducible and tested rather than silently invented.

5.4 Data model and storage

The source documents specify no runtime database. The CSV files and generated JSON in the repository act as the data store.

Recommended generated artifacts stated in the solution specification:

- `ranking.json` - per-ghat ranking and summary data.
- `evidence.json` - reading-level evidence and source context.
- Brief text and printable PDF in Hindi and English.

The exact JSON schema is not included in the supplied documents and must be recorded in the repository once defined.

5.5 Interface contracts

- The ranking view consumes generated ranking data and must display all seven defined sites.
- The evidence view consumes reading-level records and must retain year, festival, phase, value, unit, flag and source page.
- The brief is generated from a facts table and cannot introduce unverified numeric claims.
- Source receipts, if included, are generated at build time from the report page reference.
- The static host serves built artifacts over HTTPS; there is no runtime API contract in the specified design.

5.6 Architecture decisions

Decision	Selected approach	Alternative / note	Rationale from the source specification
Ranking	Deterministic Python rules	Letting an LLM choose ranks	Rules are reproducible and support trustworthy, auditable decisions
Frontend	Static HTML/JS with Tailwind or Vite + React	Streamlit is a fallback	Static deployment is shareable and requires no runtime server
Processing	Python 3 + pandas at build time	Runtime computation	Creates prepared JSON and briefs ahead of use
Runtime backend	None	REST API / WebSockets	Not required for a static data-driven product
Data storage	CSV and generated JSON in repository	Database	Fits the limited workflow and public data pack

Decision	Selected approach	Alternative / note	Rationale from the source specification
AI	Claude or Cerebras inference at build time only	Runtime AI calls	LLM is restricted to prose drafting, never numeric calculation
PDF generation	Browser print or WeasyPrint	Other PDF tooling if needed	Both are listed options; Hindi rendering must be tested early
Source page images	PyMuPDF or pdftoppm (stretch goal)	Manual linking only	Can provide receipts to original evidence
Hosting	GitHub Pages is the stated hosting option	Other static HTTPS host	A link can be opened from a phone without installation

6. Technical Approach and Core Logic

6.1 Deterministic metrics and ranking

Risk computations must be handled in Python rather than generated by an LLM. For each ghat, the specification calls for:

- Number of readings sampled.
- Number of readings breaching the applicable limit.
- Worst recorded value.
- Trend by year.
- Before/during/after sampling pattern.
- A severity/risk view and a separate evidence-strength view.

Durga records are weighted more heavily because Durga visarjan is the event being planned; Ganesh data is supporting context. The supplied specification does not state exact numeric weights or the complete formula that converts metrics into each risk tier. The implementation must define that formula and document it alongside tests.

6.2 Handling values and flags

The parser must preserve three distinct states:

1. **Numeric measurement available:** evaluate the value against a threshold only where a threshold is defined.
2. **BDL / ND:** preserve the low/non-detected meaning and original flag; do not substitute zero.
3. **Missing / not sampled:** preserve the absence of a reading; do not count it as a compliant observation or present it as a clean result.

The evidence view should retain the original printed text to support manual source checks.

6.3 Evidence-strength calculation

Evidence strength is distinct from severity. The specification describes evidence strength in terms of the number of readings and which years/phases are missing. The precise formula is not supplied. A delivered build must state how coverage is summarized, handle ambiguous site mapping explicitly, and avoid making a sparsely measured ghat appear safer due to low observation count.

6.4 Test-Me-First logic

The planned secondary ranking identifies where an additional pre-festival test is most valuable. The source specification describes the score conceptually as **severity x uncertainty**: concerning historic readings combined with missing phases or no recent table should increase the value of testing. The exact scaling/normalization is not specified and must be documented before the ranking is used operationally.

6.5 Rank-stability analysis

The planned rank-stability control recalculates the ranking under alternate choices:

- CPCB Class B versus Class C versus Class D limits.
- Durga-only data versus both festivals.
- With or without merging the Rani Kamlapati / Kamla Park records.

Ghats remaining in the top three across all tested settings may be labelled “robustly high-risk”; ghats whose ranks move should be described as assumption-dependent. Do not use these labels until the alternative scenarios have actually been calculated.

6.6 AI-assisted text and numeric audit

If an LLM is used:

- Provide the verified facts table as the source for reasons and the Hindi brief.
- Instruct it not to generate, calculate or infer new numbers.

- Extract numeric claims from the generated text and compare them with approved facts.
- Block or flag mismatched numbers and unsupported health claims.
- Show the verification result to the user.
- Keep the data/ranking pipeline functional if the LLM is unavailable, where feasible; the static ranking and evidence should not depend on runtime inference.

A visible sample badge is provided in the solution specification as an illustration ("Verified: 38 of 38 numbers, 0 health claims"). This is an example of a possible display, not a verified result for this project.

6.7 Known analytical limitations

- Total coliform is part of CPCB Class B criteria but is absent from the supplied dataset.
- There is no Class B limit in the source criteria for chromium, lead, zinc or copper; show their readings without assigning a Class B pass/fail state.
- Suspended solids are absent from the Durga 2023 and Durga 2025 tables, limiting cross-year comparisons for that parameter.
- Some ghats have incomplete sampling phases or no recent Durga table.
- The Rani Kamlapati / Kamla Park labels may refer to two places.
- Historical readings do not establish the live water condition at the time of immersion.
- The source documents do not provide the exact risk-tier formula, exact Durga-vs-Ganesh weights, or a validated forecast model. Do not imply these are supplied facts.

---

7. Technology Stack and Tools

Layer	Technology specified	Role / note
Data pipeline	Python 3	One build script, build.py
Data processing	pandas	Load, normalize and summarize CSV data
Frontend option A	Plain HTML/JavaScript with Tailwind	Lightweight static website
Frontend option B	Vite + React	Alternative static front-end approach
Frontend fallback	Streamlit	Fallback if development speed is preferred over shareability
Hosting	GitHub Pages (suggested)	Static HTTPS link for phone access
Data storage	CSV + generated JSON in repository	No runtime database specified
Optional AI	Claude or Cerebras inference, at build time only	Draft reasons and Hindi text from verified facts; never produce numbers
Verification	Python number checker + health-claim phrase list	Audit generated text and support Tamper Test
PDF	Browser print or WeasyPrint	Printable Hindi-English brief
Hindi font	Noto Sans Devanagari, bundled	Must be tested for PDF rendering
Optional page receipts	PyMuPDF or pdftoppm	Render cited report pages during build
Secrets	Environment variables	Do not commit API keys to Git

Exact dependency versions, package lock files, operating-system requirements and installation commands were not specified in the supplied documents. The repository should record them once the implementation choices are finalized.

---

8. Implementation Plan

The source is a one-day build specification for 11 October 2026. The event guide says code should be written from 10:15 AM and a public repository submitted by 5:00 PM. Since the supplied files do not include a repository status report or implementation log, the checklist below is a suggested execution sequence and must not be read as a record of completed work.

8.1 Recommended phases

Phase	Key work	Completion evidence
1. Repository and data intake	Confirm files, inspect schemas, capture environment/dependencies	Readable repository and successful CSV load
2. Trust-critical parser and metrics	Implement missing / BDL / ND handling; Class B checks; ghat metrics	Unit tests including the guide's headline counts

Phase	Key work	Completion evidence
3. Ranking and evidence model	Define documented weights / tier formula; generate severity and evidence-strength data	Reproducible ranking output with every site represented
4. Evidence and gap view	Build reading table, source-page references, missing-phase panel, ambiguous-site warning	Representative readings can be traced back to PDFs
5. Frontend and bilingual brief	Implement ranked view and Hindi-English brief/PDF	Phone-sized UI and print rendering checks pass
6. Numeric / phrase audit	Verify every numeric claim and screen forbidden health claims	Tamper demonstration flags a changed number; audit status visible
7. Final QA and release	Reproduce headline counts, spot-check source pages, smoke-test UI, README, licence and secrets	Public repository can be run by another person in five steps or fewer

8.2 Minimum viable release

The MVP should prioritize, in this order:

1. Correct data semantics and reproducible headline counts.
2. Ranking that separates severity from evidence completeness.
3. Reading-level evidence and a visible missing-data panel.
4. Short Hindi and English briefing.
5. Numeric-claim verification and a limitations statement.

The stretch features should not delay the data integrity and traceability controls.

8.3 Suggested task breakdown

Workstream	Main responsibilities	Suggested output
Data and domain research	Review CPCB criteria, source reports, naming/phase coverage and sampling caveats	Data notes and manually checked examples
Risk engine	Parsing, rules, metrics, ranks and tests	build.py, unit tests, generated ranking data
Product interface	Ranked cards, evidence drawer, gap states and responsive layout	Static dashboard
Briefing and language	Draft/review Hindi and English content, prepare PDF, check readability	Brief text/PDF and limitations wording
Validation and release	Audit numbers, inspect PDF sources, test rendering, write README, confirm licence and secrets	Acceptance checklist and public repository

These workstreams may be reassigned to a smaller or larger team. The participant guide recommends teams of four to five with technical and non-technical contributors, and expects meaningful work from every member.

8.4 Fallback plan

If time is short, retain the traceable deterministic metrics, seven-site overview, missing-data treatment, source-page evidence and a plain bilingual summary. Defer page-image receipts, shareable image cards, speech synthesis and other stretch functionality. Do not remove numeric verification or make missing data appear clean to save time.

9. Challenges, Risks and Mitigation

Risk / challenge	Why it matters	Mitigation stated or implied by the specification	Residual limitation
Missing values treated as clean	Can falsely reassure the officer	Parse missing markers explicitly; display gaps separately; test these cases	Incomplete sampling still limits conclusions
BDL/ND converted to zero	Alters the meaning of the source record	Preserve the flags and original printed value	The source does not provide a numeric detection limit for every reading
Incorrect threshold application	Produces invalid breach counts	Use explicit CPCB Class B thresholds; add boundary tests	Total coliform is not present in the data
AI invents a number	Undermines the evidence-based ranking	Keep ranks/metrics deterministic; verify generated numeric claims against the facts table	Text parsing and claim matching need careful tests
Unsupported health claims	Overstates what historical measurements show	Phrase list, human review and explicit “readings against limits only” framing	A phrase list may not detect every misleading formulation

Risk / challenge	Why it matters	Mitigation stated or implied by the specification	Residual limitation
Sparse or uneven sampling	Can make a ghat look low-risk merely due to lack of data	Separate severity and evidence strength; add missing phases/years panel; Test-Me-First prioritizes uncertainty	A gap-aware ranking still cannot replace fresh sampling
Ambiguous site matching	Could combine different physical locations	Keep Ganesh “Kamla Park kund” and Durga “Lower Lake ghat” separate and show warning	Mapping needs future field/domain verification
PDF/Hindi rendering failure	Makes the brief unusable or unreadable	Bundle Noto Sans Devanagari and test PDF rendering early	Different viewers/printers can render differently
New ghats lack historical data	Risk rank cannot be inferred from records for old sites	State historical data covers old sites; add new ghats only when data becomes available	No reliable historical rank exists until measurements are supplied
Build-time LLM unavailable	Could delay language generation	Keep deterministic metrics and structured evidence independent; use reviewed template text as fallback	Hindi drafting may need manual completion
Secrets committed to a public repo	Exposes credentials	Use environment variables and check before push	Team discipline and repository review are still required
Over-expansion beyond the user's workflow	Reduces the chance of a usable day-one product	One workflow, one primary user; defer statewide scope and portal features	Broader future integration remains out of the MVP
Historical information mistaken for live conditions	Leads to unsupported claims about current water quality	Date each reading, show sampling phase and state limitations clearly	Fresh sampling is needed for current conditions

9.1 Critical dependencies

The critical path is: correct source data and criteria -> correct parsing -> correct metrics -> auditable ranking -> verified prose/brief. A visually polished front end cannot compensate for errors in the first four steps.

9.2 Safety and compliance boundaries

The initial build uses public data only, excludes personal data and must not connect to live government systems. It must not state that a reading caused a health effect, and it must not present missing information as reassuring evidence.

10. Testing, Validation and Quality Assurance

10.1 Test strategy

Testing should cover parser/unit tests, integration from CSV to JSON, website checks, source-PDF spot checks, bilingual/PDF rendering and a user-acceptance pass on the single primary workflow.

10.2 Essential test cases

Test ID	Scenario	Expected result	Status in this document
TC-01	Run pipeline on the provided Bhopal CSV	Loader handles the expected 1,524 rows and surfaces parse errors	Not verified
TC-02	Blank, _, -, *, **, not sampled input	Stored as missing/not sampled, not clean or zero	Required test; not verified
TC-03	BDL and ND input	Status and original text are retained; values are not zeroed	Required test; not verified
TC-04	DO at 5 mg/l	Not flagged as below-5 breach	Required test; not verified
TC-05	DO below 5 mg/l	Flagged as a breach	Required test; not verified
TC-06	BOD at 3 mg/l	Not flagged as above-3 breach	Required test; not verified
TC-07	BOD above 3 mg/l	Flagged as a breach	Required test; not verified
TC-08	pH at each boundary (6.5, 8.5) and outside interval	Boundaries pass the stated interval; outside values breach it	Required test; not verified
TC-09	Headline DO statistic	Reproduces 42 of 116 readings below 5 mg/l	Required acceptance test; not verified
TC-10	Headline BOD statistic	Reproduces 100 of 112 readings above 3 mg/l	Required acceptance test; not verified
TC-11	Source-page audit	Manually checked sample of 5-10 readings agrees with report pages	Required validation; not verified

Test ID	Scenario	Expected result	Status in this document
TC-12	Sparse sampling / missing Durga year	Gap is explicit and does not imply a safe ghat	Required UI acceptance test; not verified
TC-13	Ambiguous site names	Ganesh/Durga label difference remains visible with warning	Required UI acceptance test; not verified
TC-14	LLM changes a numeric figure	Number auditor flags the mismatched claim	Required differentiator test; not verified
TC-15	Brief contains a health-effect claim	Phrase checker flags/blocks it; final brief remains scoped to readings vs limits	Required validation; not verified
TC-16	Hindi brief exported to PDF	Devanagari glyphs render cleanly and page is readable	Required rendering test; not verified
TC-17	Mobile/browser open	User can reach ranking, evidence and brief without install or login	Required usability test; not verified
TC-18	Ranking stability scenario	Results recalculate according to each documented assumption set	Only applicable if the differentiator is implemented; not verified

10.3 Acceptance criteria

A release should not be described as ready until the headline counts reproduce, a small set of values has been checked against the original PDFs, missing and BDL states remain distinct throughout the pipeline, numeric claims pass the facts-table checker, unsupported health claims are absent, all seven ghat cards work, the ambiguity warning is present and the bilingual brief renders correctly.

10.4 Known limitations

The source dataset lacks total coliform for Class B, has uneven time/phase coverage, has uncertain site matching for the Rani Kamlapati / Kamla Park labels, and does not provide Class B criteria for the metals. The supplied documents do not establish that the current code or deployment has passed the tests above.

---

11. Deployment, Operations and Maintenance

11.1 Deployment model

The recommended model is a public static website over HTTPS, with GitHub Pages named as the main option. An officer should be able to open the link on a phone without installing an app or signing in. No live backend or database is needed at runtime.

11.2 Build and release workflow

1. Confirm the source data and criteria files in the repository.
2. Configure any build-time inference key as an environment variable; do not store it in source control.
3. Run `python build.py` to generate ranking, evidence and briefing artifacts.
4. Run automated tests and manual PDF spot checks; review the numeric-audit result and Hindi/PDF rendering.
5. Publish the static files and check the deployed link from a phone-sized browser.

The above is a proposed concise operating procedure consistent with the specified architecture; exact commands beyond `python build.py` must be confirmed in the actual repository.

11.3 Generated artifacts

- `ranking.json`
- `evidence.json`
- Bilingual brief text
- Printable bilingual PDF
- Optional source-page images and evidence-linked QR codes if the receipts stretch goal is implemented
- Optional bilingual image cards if Pocket Briefing is implemented

11.4 Monitoring and logging

Because runtime processing is static, the main operational checks are build success, output validity, failed verification claims, source data changes and deployed-file integrity. Logs should not contain API keys or other secrets. A visible data/report date helps users understand the age of the evidence.

11.5 Maintenance plan

For a new report release or new sampling event, add the source data and PDF, verify site names/phases/source-page mappings, rerun the build and tests, review changes in rank and evidence coverage, regenerate the brief, then publish. Do not merge new ghats into the seven-site ranking until there is suitable data and site mapping.

11.6 Recovery and continuity

Retain the source CSVs, report PDFs, code and generated artifacts in the repository so the static output can be rebuilt. If the LLM provider is unavailable, the numeric ranking and evidence should remain available; use a reviewed template or manual prose for the brief rather than inventing content. This fallback behavior should be tested.

---

12. Impact, Value and Differentiation

12.1 Expected impact

The expected value, not yet a measured outcome, is that the BMC officer can review historic concerns and gaps faster and base follow-up decisions on traceable evidence rather than reconstructing information manually from multiple PDFs.

Potential benefits include:

- **Clearer priorities:** a ranked view brings higher-concern sites into view.
- **Better testing decisions:** gaps become explicit and can drive a “test first” recommendation.
- **More transparent reasoning:** every ranking reason can be inspected against readings and limits.
- **More usable communication:** a short Hindi and English brief can be forwarded to relevant teams.
- **Reduced risk of misleading output:** deterministic calculations and numeric verification constrain AI-generated explanations.
- **Better institutional handover:** public, readable code and reproducible artifacts support future maintenance.

No measured time savings, cost savings or environmental improvement are claimed in the source material.

12.2 Unique value proposition

> *For the BMC officer planning Durga visarjan, Visarjan Ghat Watch organizes historical MPPCB readings into a traceable, bilingual risk briefing that shows both where readings breach chosen limits and where the evidence is incomplete, helping the officer prioritize follow-up without treating missing data as clean.*

12.3 Differentiation

Alternative / baseline	Visarjan Ghat Watch difference	Evidence needed to support the claim
Manual review of individual PDFs	Consolidated cross-year, per-ghat view	Reproducible output from the supplied CSV and report checks
Single overall score	Severity and evidence strength shown separately	UI review and tests with sparse vs well-sampled sites
AI-generated narrative without audit	Numbers must match an approved facts table	Numeric checker test and tamper demonstration
Generic dashboard without data caveats	Missing phases, years and site ambiguity are explicit	Acceptance tests for known data gaps
Long technical report	Short Hindi-English summary for a defined officer workflow	Readability review with intended users

These are intended differentiators. Superiority over other tools has not been independently benchmarked in the supplied files.

12.4 Limitations of the claimed advantage

The solution is only as useful as the quality, coverage and correct interpretation of its historical data. It is not a real-time monitoring platform, does not include all Class B parameters, and cannot establish present-day conditions at a ghat without new measurements. The rankings may change under different criteria or data-merging assumptions; the rank-stability control is intended to make that dependence visible.

---

13. Feasibility, Viability and Sustainability

13.1 Technical feasibility

The proposed implementation uses a compact batch pipeline (Python + pandas), generated JSON and a static website, with no runtime database or backend. This architecture is aligned with the narrow workflow and the event's one-day build constraint. Feasibility still depends on correctly implementing and testing parser semantics, rank logic, numeric auditing and Hindi PDF rendering.

13.2 Economic feasibility

The supplied documents do not provide a budget, hosting cost, inference cost or ongoing maintenance estimate. Static hosting and locally generated artifacts avoid the need for a continuously running application server, but build-time inference may incur usage costs depending on the provider and plan. Any cost claim should be made only after selecting the provider and estimating usage.

13.3 Operational viability

The intended user is narrowly defined, and the product is designed to open from a phone without installation or login. A short ranked view, evidence drawer and one-page brief fit the stated workflow. Adoption and actual usefulness still need to be confirmed by the BMC user or a domain expert; no user-acceptance result is included in the supplied documents.

13.4 Long-term sustainability

Maintenance depends on adding new public sampling data, validating site names and source pages, maintaining thresholds and methodology documentation, and rerunning the tests with each data update. The project should keep source data, calculation logic and generated outputs in a reproducible repository. Extension to new ghats or statewide data needs a separate data-quality pass.

13.5 Commercial viability

A commercial model is not specified and should not be forced onto this public-interest, open-source decision-support project. The guide states that the repository is to be offered to the Government of Madhya Pradesh through MPSeDC and should include a licence, with MIT suggested unless the organizers advise otherwise.

---

14. Team, Responsibilities and Governance

The participant guide recommends teams of four to five with a mix of technical and non-technical contributors. Names and individual assignments are not included in the supplied files, so the following is a suggested allocation rather than a record of the actual team.

Role / workstream	Responsibilities	Evidence of completion
Project coordination	Keep the build focused on the BMC officer's one workflow; track acceptance criteria	Agreed scope and status board
Data and domain research	Review CPCB criteria, inspect source reports, verify naming and phase gaps	Source notes and manually checked sample rows
Data/risk engineering	Implement parsing, metrics, ranking and automated tests	Reproducible build and test results
Interface engineering	Create ranking cards, evidence panel, gaps display and responsive layout	Working static page
Briefing / validation	Review Hindi and English phrasing, PDF output, number checker and limitations	Approved brief and verification report
Release / documentation	README, run steps, licence, dependency notes and secret scan	Public repo with runnable instructions

Some roles can be combined. Every team member should have a clear deliverable. The project guide specifies that the repository be public and readable, and that the README explain the team, track, primary user, how to run the tool in five steps or fewer, what works and does not work, any extra data sources, and the licence.

14.1 Ownership and access

The actual project owner, team member names and intellectual-property arrangements are not established in the supplied files. These should be recorded in the repository README. Use public data only for the event build, do not include personal data, and keep API keys outside source control.

---

15. Future Roadmap and Expansion

15.1 Initial release

Focus on the seven Bhopal ghats, three supported parameter thresholds, evidence traceability, visible sampling gaps, the bilingual brief and numeric auditing. Verify all outputs before adding scope.

15.2 Near-term improvements

- Test-Me-First ranking to prioritize additional testing before the event.
- Tamper Test UI and visible numeric verification status.
- Rank-stability comparisons for alternate CPCB classes, festival weighting and site-merge assumptions.
- Clickable PDF receipts and evidence-linked QR code, if time allows.
- WhatsApp-friendly bilingual image cards and Hindi speech playback, if time allows.

15.3 Longer-term enhancements stated in the source documents

- Expand to the 21,600-row statewide dataset after reviewing rows flagged `needs_check` and standardizing site identities.
- Add total coliform only when suitable data is available; it is a Class B criterion absent from the current dataset.
- Add the four new BMC ghats when there are appropriate monitoring records for them.
- Add Upper and Lower Lake reports for context.
- Consider integration with live monitoring after the event, as a future phase rather than part of the current MVP.

15.4 Upgrade policy

Each extension should preserve backward-compatible source fields, original printed values, source-page provenance and explicit data-status semantics. Recompute outputs after changing thresholds or weighting, rerun all headline and numeric-audit checks, review ranking changes, and record the method/version used to produce the briefing.

---

16. Documentation, References and Glossary

16.1 Primary project sources

1. **Visarjan Ghat Watch: Solution Architecture & Project Specification.** Supplied PDF, dated 11 October 2026. Defines the MVP, feature priorities, architecture, data handling, numeric verification and stated exclusions.
2. **Track 2: Visarjan Ghat Watch - Participant Guide.** Supplied PDF, event date 11 October 2026. Defines the problem, dataset inventory, site list, CPCB criteria, event constraints and assessment criteria.
3. **Track repository:** https://github.com/Laaaaksh/bhopal-impact-lab-2026 - the guide states that files are under `tracks/02-visarjan-ghat-watch` and that the same files are in the Drive folder.

16.2 Public references cited by the guide

- **CPCB water-quality criteria:** https://cpcb.gov.in/water-quality-criteria/
- **MPPCB monitoring reports:** the guide lists the 2023, 2024 and 2025 Ganesh/Durga immersion reports on MPPCB's website. Use the supplied PDFs and the corresponding repository copies for reproducible validation.
- **NGT ban / new ghats context:** the participant guide cites a Free Press Journal article dated 26 February 2026. The exact article URL printed in the guide is: Free Press Journal article.
- **Durga visarjan date:** the participant guide cites Drik Panchang for 20 October 2026: Drik Panchang.

The external references above are reproduced from the supplied participant guide; this document does not independently validate whether those pages remain unchanged.

16.3 Glossary

Term	Meaning in this project
BMC	Bhopal Municipal Corporation
MPPCB	Madhya Pradesh Pollution Control Board
CPCB	Central Pollution Control Board
NGT	National Green Tribunal
DO	Dissolved oxygen
BOD	Biochemical oxygen demand
BDL	Below detection limit; not zero and not missing

Term	Meaning in this project
ND	Not detected; treated as very low, not zero and not missing per the guide
CSV	Comma-separated values data file
JSON	JavaScript Object Notation; format for generated ranking/evidence files
LLM	Large language model; used only for prose drafting in the specified design
MVP	Minimum viable product
Evidence strength	Coverage/availability of readings by count, year and phase, separate from severity
Severity	How strongly available readings breach the selected criteria

16.4 Source integrity labels

Use these labels consistently in future documents and presentations:

- **Source-stated:** explicitly provided in the solution specification or participant guide.
- **Derived requirement:** a check or implementation behavior implied by a stated feature.
- **Implementation choice:** a formula, schema or technical detail selected by the builders; document it and test it.
- **Verified in build:** supported by recorded automated/manual test output from the actual repository.
- **Not yet verified:** expected behavior without evidence that the current build passes.

17. Open Questions and Action Items

17.1 High-priority decisions

Item	Why it matters	Suggested owner	Status
Confirm exact repository layout and all input file paths	Avoid build failures from filename/path assumptions	Data/risk engineer	Open / not confirmed
Define exact risk-tier formula and Durga-vs-Ganesh weights	Ranking must be reproducible and explainable	Risk-engine owner	Not specified in supplied documents
Define evidence-strength calculation	Data coverage must be visible without being confused with severity	Risk-engine owner	Not specified in supplied documents
Define Test-Me-First normalization for severity x uncertainty	Secondary recommendation ranking needs a reproducible formula	Risk-engine owner	Concept specified; formula not provided
Confirm whether Rani Kamlapati / Kamla Park records remain separate in default ranking	Ambiguous site identity can alter ranks	Data/domain owner	Warning required; final default behavior needs documentation
Decide static frontend option	Determines project scaffolding and build process	Frontend owner	Options listed; final choice not established
Define JSON schema	Frontend and generated outputs need stable contracts	Data/risk + frontend owners	Not supplied
Verify headline counts and sample report readings	Establishes trust in the underlying facts	QA/data owner	Required; results not supplied
Test the actual numeric auditor against generated text	Prevent ungrounded numbers from reaching user	QA/verification owner	Required; results not supplied
Test Hindi rendering and PDF output	Makes the bilingual brief usable	Briefing owner	Required; results not supplied
Confirm repository licence and secret scan	Required for readable public handover	Release owner	Guide recommends MIT unless organizers specify otherwise
Confirm implementation/deployment status	The supplied PDFs do not report a final build status	Project lead	Open

17.2 Readiness checklist

- [ ] Input dataset and source PDFs are present at documented paths.
- [ ] Missing markers are treated as missing, distinct from BDL and ND.
- [ ] CPCB Class B thresholds are encoded and documented.
- [ ] DO and BOD headline unit tests reproduce 42/116 and 100/112 respectively.
- [ ] Five to ten readings are manually checked against the cited PDF pages.
- [ ] Seven ghats appear in the ranking view.
- [ ] Severity and evidence strength are separate in the interface.
- [ ] Missing years/phases and ambiguous site naming are visibly disclosed.
- [ ] Each generated numeric claim is verified against the facts table.
- [ ] Health-claim screening has been tested and the limitations note is visible.
- [ ] Hindi and English brief are readable in the browser and printable PDF.
- [ ] README includes team, track, target user, run steps, limitations and any extra data sources.
- [ ] Licence file is present and appropriate.
- [ ] No secrets or API keys are committed.
- [ ] Public static deployment works on a phone-sized browser.
- [ ] Remaining limitations and unimplemented features are accurately documented.

17.3 Handover rule

A feature should be labelled **implemented** only when its code exists and a relevant check has passed. A feature in the solution specification but not demonstrated by the repository should remain **planned** or **not verified**. Ranking results should always carry the criteria/method context needed to understand them, and data gaps must remain explicit.